End of Line Report

Program Name:	Firefly Motor	Motor Code:	ML2A05-C05
Motor Model:	ML2B	Serial no:	A260506A000056

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Datasheet

Parameter	Golden Sample	DUT	Deviation
Model	ML2A05B-51V_P39&P40	A260506A000056	-
Nominal DC Voltage	52	52	-
Peak Power	10.80	11.29	-
Peak Torque	63.64	65.32	-
Peak Current - AC	356.46	360	-
Motor Max Speed	7200	7200	-
Document Ref.	EOL - ML2A05B-51V_P39&P40		

Back-EMF

Speed	AVG	DUT	Deviation
500	4.44	4.48	0.85%
1000	8.89	8.81	-0.87%
1500	13.33	13.17	-1.19%
2000	17.78	17.56	-1.20%
3000	26.67	26.36	-1.15%
4000	35.55	35.16	-1.10%
5000	44.44	43.95	-1.10%
5500	48.89	48.37	-1.07%

Validation Checklist

Parameter	Compliance
Back EMF	ok
Peak Power	ok
Peak Torque	ok
Characterstics Curves	ok
Average Efficiency	ok

Comparison - Torque Specs

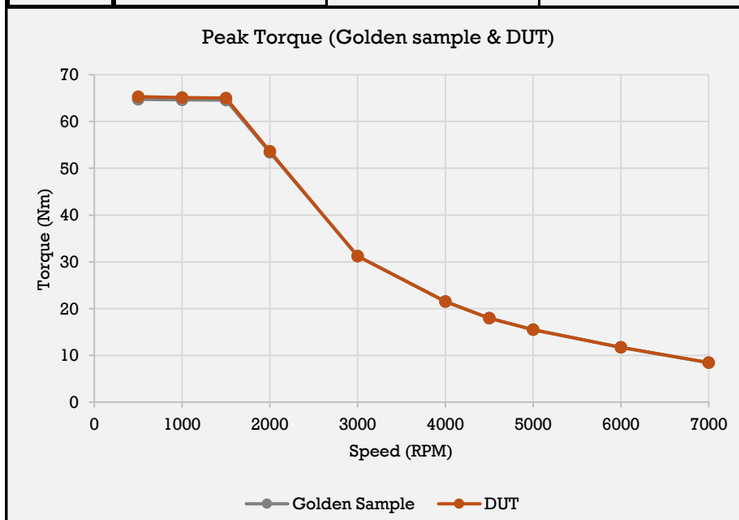
Speed	Golden Sample	DUT	Deviation
500	64.76	65.32	0.86%
1000	64.65	65.14	0.77%
1500	64.60	65.01	0.63%
2000	53.39	53.65	0.49%
3000	31.24	31.21	-0.08%
4000	21.53	21.46	-0.33%
4500	17.98	17.94	-0.22%
5000	15.49	15.48	-0.06%
6000	11.71	11.69	-0.13%
7000	8.43	8.45	0.24%

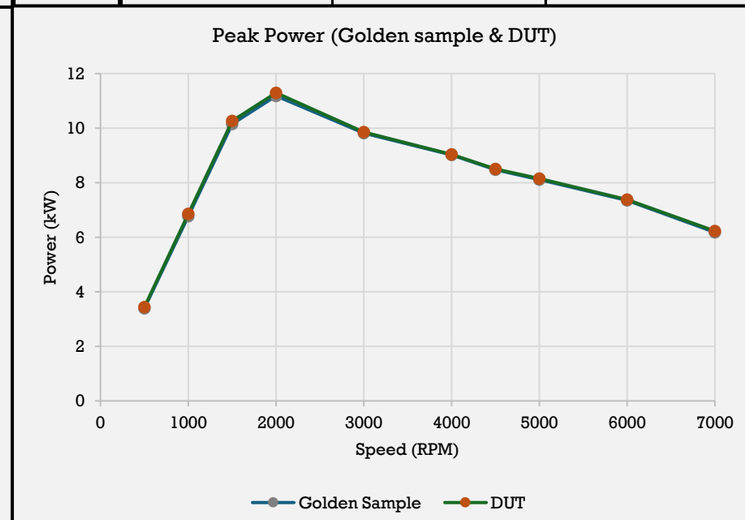
Comparison - Power Specs

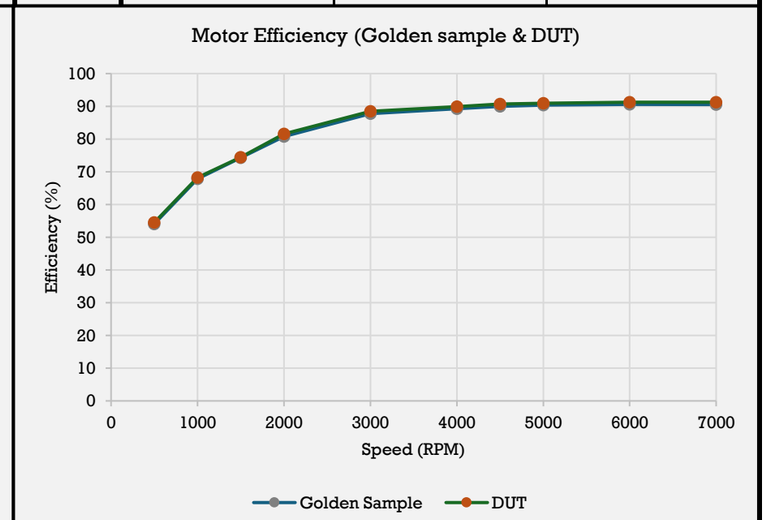
Speed	Golden Sample	DUT	Deviation
500	3.39	3.44	1.40%
1000	6.77	6.86	1.30%
1500	10.15	10.26	1.16%
2000	11.18	11.29	1.02%
3000	9.81	9.86	0.45%
4000	9.02	9.04	0.20%
4500	8.47	8.50	0.30%
5000	8.11	8.15	0.46%
6000	7.35	7.38	0.40%
7000	6.18	6.23	0.76%

Comparison - Group Efficiency Specs

Speed	Golden Sample	DUT	Diffrence
500	54.06	54.49	0.79%
1000	67.82	68.25	0.64%
1500	74.27	74.43	0.22%
2000	80.78	81.55	0.96%
3000	87.75	88.49	0.84%
4000	89.28	89.93	0.73%
4500	90.00	90.69	0.77%
5000	90.37	90.92	0.62%
6000	90.58	91.24	0.73%
7000	90.51	91.23	0.80%







Integration Details	sin min	1162.00	cos min	1162.00	sin max	2979.00	cos max	2973.00	sinP-P	1817.00	cosP-P	1811.00	Offset	766
Electrical PDI	Leakage current(mA)	7.71			Insulation Resistance(MOhms)	9999			RESISTANCE(mOhms)	8.87				

END OF LINE RESULT	PASS	Tested by	Approval
		HRN - 06 MAY 2026	This is a system-approved document; signatures are not required.